

AI & ML by Aditya Jain - Lecture 1.1

```
#Tentative Course Plan:
# 1. Python Essentials --> 12
# 2. Python for ML & AI --> 12
# 3. SQL for ML & AI --> Structured Query Language --> 18
# 4. Numpy --> Python Library for Numerical Python --> 6
# 5. EDA using Pandas --> Efficiently deal with Tabular data using Python --> 12
# 6. Data Visualization using Python --> Matplotlib, Seaborn --> 6
# 7. Probability & Statistics --> 12
# 8. Classical ML --> Classification, Regression --> 12
# 9. Deep Learning --> Neural Networks. --> 12
# 10. GenAI & Prompt Engineering --> LLM, Frameworks for Prompt Engineering. --> 16
# 11. RAG, Agents, MCP --> 16
# 12. Project & Deployment --> 4-5
```

Important Terminologies

```
# Algorithm--> step by step instructions to solve any problem.

# Programming --> is a way of writing these instructions in a language that the computer understands.

# Programming Languages --> C, C++, Python, Java, JavaScript...

# Why Python? -->
# 1. Easy to understand --> programmer friendly
# 2. Rich set of Libraries--> AI, ML, Web Dev, Automation...

# Hello world program --> C++ and Python

# C++:
# #include<iostream>
# using namespace std;

# int main()
# {
#     cout << "Hello World!";
# }

# Python:
# print("Hello World!")

# "Apple is Aditya Jain is eating"
# -----
# "Aditya Jain is eating an Apple"

# What is Syntax?
# Grammatical rules for that particular programming language

# Why Google Colab Notebook?
# 1. No installation
# 2. Free
# 3. Runs in Browser
# 4. Used across industry
```

Basics of Python Programming

```
print("Hello World!")
```

```
Hello World!
```

```
print("Hello World!")
```

Hello World!

```
print("Hello World!")
```

Hello World!

```
print("Welcome to Python")
```

Welcome to Python

```
print(Welcome to Python)
```

```
File "/tmp/ipykernel_8021/1871024563.py", line 1
  print(Welcome to Python)
      ^
SyntaxError: invalid syntax. Perhaps you forgot a comma?
```

Next steps: [Explain error](#)

```
print("Welcome to Python")
```

Welcome to Python

```
print("A")
print("B")
print("C")
```

A
B
C

```
print("A" , "B" , "C")
```

A B C

```
print(100)
```

100

```
print(3+5)
```

8

```
print(5-3)
```

2

```
print(4*2)
```

8

```
print(4/2)
```

2.0

```
print("3+2")
```

3+2

```
print(3+2)
```

5

Quizes

1. Which of the following is a common use of Python?

```
# A. AI & Machine Learning

# B. Data Science

# C. Automation

# D. All of the Above

# Ans: D
```

```
# 2. What is Google Colab?

# A. A social media platform

# B. An online Python coding environment

# C. A gaming application

# D. A database

# Ans: B
```

```
What will be the output?

print("Hello World")

A.

Hello World

B.

"Hello World"

C.

print("Hello World")

D. Error

Ans: A
```



HW:

```
# 1. What will be printed?

# print(10)

# A. Ten

# B. "10"

# C. 10

# D. Error
```

```
# 2. Guess the output:

# print(5 + 3)

# Output?

# A. 53

# B. 8

# C. 5+3

# D. Error
```

```
print("Thanks! Hope it was useful : ")
```

```
Thanks! Hope it was useful : )
```

Start coding or [generate](#) with AI.